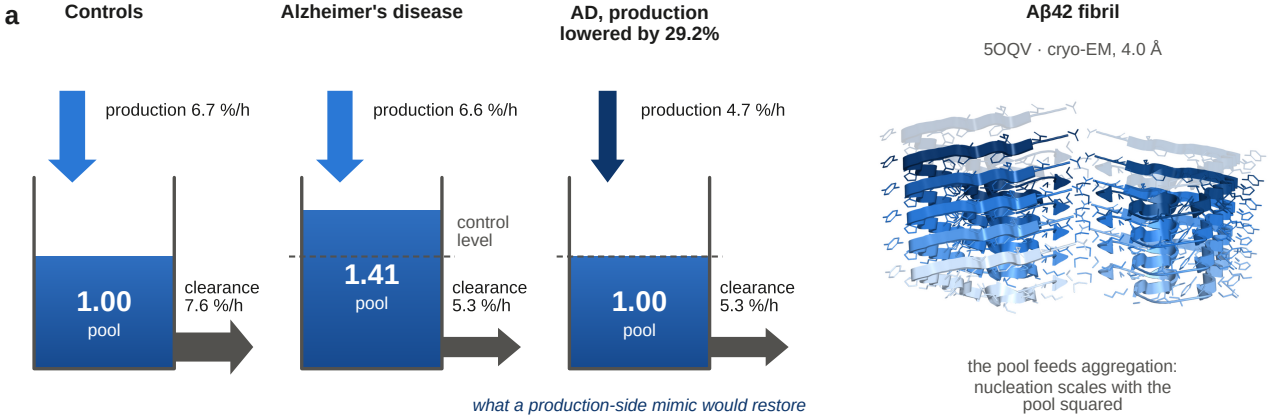
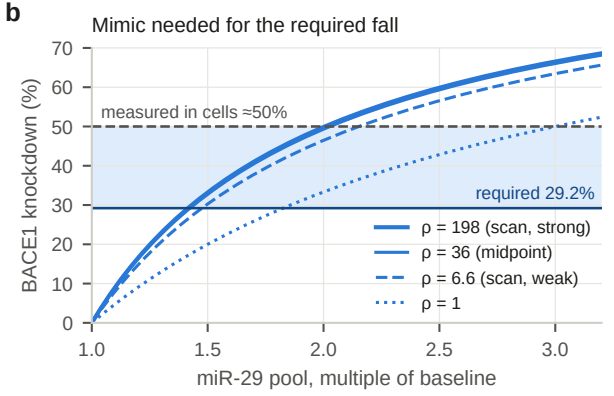
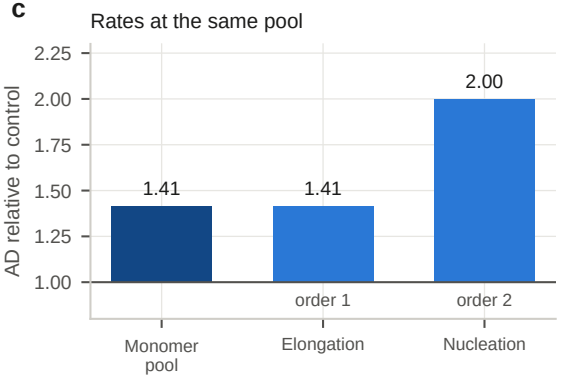

a
Controls
Alzheimer's disease
AD, production lowered by 29.2%
Aβ42 fibril
5OQV · cryo-EM, 4.0 Å
production 6.7 %/h
1.00
pool
clearance
7.6 %/h
production 6.6 %/h
control level
1.41
pool
clearance
5.3 %/h
production 4.7 %/h
1.00
pool
clearance
5.3 %/h
what a production-side mimic would restore
the pool feeds aggregation:
nucleation scales with the
pool squared


b
Mimic needed for the required fall
BACE1 knockdown (%)
measured in cells ≈50%
required 29.2%
ρ = 198 (scan, strong)
ρ = 36 (midpoint)
ρ = 6.6 (scan, weak)
ρ = 1
miR-29 pool, multiple of baseline


c
Rates at the same pool
AD relative to control
1.41
1.41
2.00
order 1
order 2
Monomer pool
Elongation
Nucleation